

Problem

Find the Fourier transforms of the following functions:

(a) $f(t) = \delta(t + 3) - \delta(t - 3)$

(b) $f(t) = \int_{-\infty}^{\infty} 2\delta(t - 1) dt$

(c) $f(t) = \delta(3t) - \delta'(2t)$

Step-by-step solution

Step 1 of 3

(A) $F(\omega) = e^{3j\omega} - e^{-3j\omega} = 2j \sin 3\omega$

Step 2 of 3

(B) Let $g(t) = 2\delta(t - 1)$, $G(\omega) = 2e^{-j\omega}$

$$\begin{aligned} F(\omega) &= F\left(\int_{-\infty}^t g(t) dt\right) \\ &= \frac{G(\omega)}{j\omega} + \pi F(0)\delta(\omega) \\ &= \frac{2e^{-j\omega}}{j\omega} + 2\pi\delta(-1)\delta(\omega) \\ &= \frac{2e^{-j\omega}}{j\omega} \end{aligned}$$

Step 3 of 3

(C) $F[\delta(2t)] = \frac{1}{2} \cdot 1$

$$F(\omega) = \frac{1}{3} \cdot 1 - \frac{1}{2} j\omega = \frac{1}{3} - \frac{j\omega}{2}$$